



North South University

Department of Mathematics and Physics

MAT350: Engineering Mathematics

Course Name: Engineering Mathematics
Course Code: MAT 350
Credit Hours : 3 Credits
Pre-requisite : MAT 130
Term : FALL 2026

Course Coordinator : Dr. Mohammad Sahadet Hossain

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Office Time : ST 2pm-4pm, MW 2pm-4pm, and by appointment

Course Short Description: This course is intended for Engineering students who require a working knowledge of differential equations; included are techniques and applications of ordinary differential equations in engineering problems with some elements of scientific computing.

Course Objectives

1. To classify the different types of differential equations and find the appropriate analytical tools and techniques for finding the solutions of the first order and the second order ordinary differential equations.
2. To create and analyze mathematical models using the first order and the second order ordinary differential equations.
3. To demonstrate students' understanding of how physical phenomena are modeled by system of differential equations and investigate the solution methods.
4. To develop the ability to apply Fourier series and Fourier Integrals to significant applied problems.

Course Learning Outcomes: Upon successful completion of this course, students will be able to:

(CO-1) Classify the type of a given differential equation and find the appropriate analytical techniques for finding the solutions of the first order and the second order ordinary differential equations.

(CO-2) Formulate and analyze mathematical models using the first order and the second order ordinary differential equations.

(CO-3) Solve linear differential equations using different tools, like the Laplace transform technique, power series method; and identify their applications.

(CO-4) Demonstrate their understanding of how physical phenomena are modeled by system of differential equations and investigate the solution methods.

(CO-5) Develop the ability to apply Fourier series and Fourier Integrals to significant applied problems.

**Course
Contents:**

1. **First Order Ordinary Differential Equations:** Introduction to Ordinary Differential Equations (ODEs), mathematical modeling with ODEs, Separable and Exact ODE, Linear ODE, Bernoulli equation.
2. **Second-Order Linear ODEs:** Homogeneous Linear ODEs of Second Order, Homogeneous Linear ODEs with Constant Coefficients, Euler–Cauchy Equations, Existence and Uniqueness of Solutions. Wronskian, Non-homogeneous ODEs, Modeling: Mass–Spring System, Electric Circuits, Resonance, Solution by Variation of Parameters.
3. **Systems of ODE:** System of ODEs, Phase plane method, Nonhomogeneous Linear Systems of ODEs.
4. **Series Solutions of ODEs:** Power series method, Extended Power Series Method, Bessel’s Equation. Bessel Functions and general solution.
5. **Laplace Transformation:** Laplace Transformation and its inverse, linearity and shifting, Laplace transformations of derivatives and integrals, Initial Value Problems, unit step function, delta function and t-shifting.
6. **Fourier Series and Fourier Functions:** Periodic function and Fourier Series, Fourier coefficients and applications. Even and odd functions, Half range expression, Fourier integrals and transforms.

Mapping of Course Outcomes

#	Course Outcomes (CO)	Bloom’s taxonomy domain/level (C: Cognitive P: Psychomotor A: Affective)	Delivery methods and activities	Assessment tools
CO-1	Classify the type of a given differential equation and find the appropriate analytical techniques for finding the solutions of the first order and the second order ordinary differential equations.	C2	Lecture, Video Discussion	Quiz, Assignment
CO-2	Apply and analyze mathematical models using the first order and the second order ordinary differential equations.	C3, C4, P2	Lecture, in-class group discussion, Videos	In class assessment, Midterm exam, Assignment
CO-3	Solve linear differential equations using different tools, like the Laplace transform technique, power series method; and identify their applications.	C2, C3, P2	Lecture, Discussion	Class work, Quiz, Assignment, Midterm Exam, Final Exam
CO-4	Demonstrate their understanding of how physical phenomena are modeled by system of differential equations and investigate the solution methods.	C4, P2	Lecture, Video, Discussion	Demonstration, Quiz, Assignment, Final Exam
CO-5	Develop the ability to apply Fourier series and Fourier Integrals to significant applied problems.	C2, C3, P2	Lecture Video Demonstration	Assignment, Final Exam

Note: C2, C3, C4, and P2 indicate different subdomains of Bloom’s Taxonomy. Please visit departmental website for details.

Marks Distribution:

Attendance	5%
Regular Quizzes	20% (Min 3 quizzes).
Mid-term	30%
Final Exam	35%
Assignment	5% (Two Assignments)
Quiz on Assignment (Syllabus: the syllabus of the Final exam)	5% (as per the directives of the department, one quiz will be based on the syllabus of the 2 nd Assignment)
Total Marks	100%

Grading**Policies:**

As per NSU Grading Policy

Important**Dates:**

Midterm exam

TBA

Final Exam

According to NSU Schedule.

Reference Books:

1. A First Course in Differential Equations with Modeling and Applications, (10th Edition), Author- Dennis G. Zill.
2. Advanced Engineering Mathematics (10th Edition)- Author: Erwin Kreyszig

Special Instructions:

- (a) Submit the assignments in recommended date. **No late submission will be accepted.** Make a photocopy of your assignment before submission. No Makeup of Quiz is possible in any circumstances. We consider Best 2 Quiz. So if you miss a quiz, fight for the next one.
- (b) Exam and Makeup policy is given in the last page of this course outline
- (c) **A late present means you come to the class within 10 minutes the class starts.** You are automatically **absent after 15 minutes delay (sharp)** and not allowed to the class room.
- (d) If you are a **probation student/retake student**, I would like to have you in 24 classes (**20 classes are must**)

Lesson Plan:

Lesson	Topics	Learning Activities	Assessment tools	Learning Outcome
I	First Order Ordinary Differential Equations: Introduction to Ordinary Differential Equations (ODEs), mathematical modeling with ODEs, Separable ODEs.	Lecture	Discussions Mid term	CO-1
II	First Order ODEs- Separable ODEs Separable ODEs with modeling and applications.	Individual Assignment	Quiz 1	CO-1
III	First Order ODEs- Exact ODEs Exact ODE, Linear ODEs	Lecture Group Discussion	Discussions Quiz 1 Mid term	CO-1
IV	First Order Linear ODE: Linear ODE with modeling and applications.	Lecture Discussion	Quiz 1 Mid term	CO-1
V	Bernoulli equations with modeling and applications.			
VI	Second-Order Linear ODEs: Homogeneous Linear ODEs of Second Order: Types and Solution methods.	Lecture	Mid term	CO-1

VII	Second-Order Linear ODEs: Homogeneous Linear ODEs with Constant Coefficients: Superposition principle and inverse operator method.	Lecture Assignment	Quiz 2 Mid term	CO-1
VIII	Second-Order Linear ODEs: Homogeneous Linear ODEs with Constant Coefficients: Shift exponents and variation of parameters method	Lecture	Quiz 2 Mid term	CO-1
IX	Second-Order Linear ODEs: Modelling Modelling: Mass–Spring System without/with damper	Discussion Lecture Assignment	Mid term	CO-2
X	Second-Order Linear ODEs: Modelling Modelling: Mass–Spring System without/with damper	Lecture Assignment Prepare for Mid I	Mid term	CO-2
XI	Non-homogeneous ODEs: Cauchy Euler Equation and Variation of parameters	Lecture	Mid term	CO-1
XII	Midterm Exam (Lesson 1 to lesson X)			
XII	Systems of ODEs System of ODEs: Homogeneous system	Lecture assignment	Quiz 3	CO-4
XIII	Systems of ODEs Nonhomogeneous Linear Systems of ODEs.	Lecture assignment	Quiz 3	CO-4
XIV	Systems of ODEs Nonhomogeneous Linear Systems of ODEs.	Lecture assignment	Quiz 3 Final Exam	CO-4
XV	Laplace Transformation Laplace Transformation and its inverse.	Lecture assignment	Final Exam	CO-3
XVI	Laplace Transformation linearity and shifting, Laplace transformations of derivatives and integrals,	Lecture assignment	Final Exam	CO-3
XVII	Laplace Transformation Initial Value Problems, unit step function, delta function and t-shifting.	Lecture Assignment Prepare for Mid II	Final Exam	CO-3 CO-2
XVIII	Fourier Series and Fourier Functions Periodic function and Fourier Series, Fourier coefficients and applications. Even and odd functions.	Lecture assignment	Quiz 4 (optional) Final Exam	CO-5
XX	Fourier Series and Fourier Functions Fourier coefficients and applications. Even and odd functions, Half range expression	Lecture assignment	Quiz 4 (optional) Final Exam	CO-5
XXI	Fourier Series and Fourier Functions Fourier integrals and transforms	Lecture assignment	Quiz 4 (optional) Final Exam	CO-5
XXII	Series Solutions of ODEs Power series method-about ordinary point	Lecture assignment	Final exam	CO-3
XXIII	Series Solutions of ODEs Power series method-about singular point	Lecture assignment	Final exam	CO-3
XXIV	Preparation for final	Q and A session	Final exam	
Final Exam				

Classroom Rules of Conduct: Please Refer to NSU Student Handbook, Sections: “Disciplinary Actions” and “Procedures and Guidelines”.

Exams & Make Up Policy

If a student misses the Midterm exam and/or Final exam due to the circumstances beyond their control (valid documents/medical reports/other papers are required) and informed beforehand (as early as possible), reasonable arrangement may be considered.

Please note the retake exam questions are generally a bit tricky and critical compare to the regular exam questions.
Students will get the opportunity to see/recheck their midterm and Final exam scripts.